

MAL1471 2026 Final Project

Phase 2 Assignment Brief

Author: Emmanuel Kalunga

Preamble

You have described your organisation and mapped how work flows through it. Phase 2 asks you to look at that same environment through a different lens: data. Every step in your value stream generates, consumes, or transforms information. Some of that information is captured and stored. Some is observed but never recorded. Some is not even noticed. This phase asks you to map that data landscape: what the organisation knows, how it knows it, and what it is missing.

Read carefully before beginning.

Your Phase 3 problem framing will be built on the foundation of this analysis. A shallow or generic data landscape will produce a weak Phase 3. Every data source and gap you identify must be traceable to a specific step or handover point in your Phase 1b value stream map. Do not invent data that was not visible there.

Assignment

Produce a data landscape analysis of your organisation or business unit. Your analysis has two components:

- Data inventory: what data does the organisation currently capture? For each source, document what it contains, where it is stored, who owns it, in what format, and your assessment of its completeness and quality.
- Data gap analysis: what information should the organisation be capturing but is not? For each gap, explain concretely what ML problem it would enable if it were collected.

Minimum deliverables

- A data inventory covering the major data sources visible in your Phase 1b value stream map (minimum 3 sources). For each: what it contains, storage format, owner, and a brief quality assessment.
- A data gap analysis: at least 2 gaps i.e. information the organisation should capture but does not. For each gap, explain what ML problem it would enable.
- A data opportunity table: a structured summary mapping each data source or gap to a potential ML application (e.g. classification, regression, clustering). Use the template below.
- Prioritisation: which data source or gap represents the most valuable ML opportunity, and why?

Scope

Focus on data that is visible from your Phase 1b value stream map. You are not expected to access, download, or analyse actual data in this phase. This is an inventory and gap analysis, not a modelling exercise.

What good looks like

- Is specific, not generic.
 - Weak: "the company has some customer data."
 - Strong: "the credit department maintains a PostgreSQL database of 50,000 loan applications since 2018, with 23 fields including applicant income, requested amount, employment sector, and repayment outcome. The repayment outcome field is populated for 94% of records, making this a near-complete supervised dataset."
- Gap analysis is actionable: not "they should collect more data" but "the operations team records delivery times but does not capture driver ID, vehicle type, or weather conditions at dispatch. Adding these three fields would enable a delivery delay prediction model."
- Demonstrates understanding of the supervised/unsupervised distinction: a classification or regression problem requires a label; a clustering problem does not. The opportunity table should reflect this.
- Prioritisation is reasoned: considers ML value, data availability, quality, and the feasibility of collection within the Phase 3 timeframe.

Timeline

Phase	Publication	Submission Deadline	Status
Phase 1(a)	25 May 2026	27 May 2026 (Wed)	SUBMITTED
Phase 1(b)	25 May 2026	30 May 2026 (Sat)	SUBMITTED
Phase 2	01 Jun 2026	06 Jun 2026 (Sat)	CURRENT
Phase 3	08 Jun 2026	20 Jun 2026	Upcoming

Submit as a single PDF or Word file Saturday 06 June 2026 or before.

Filename: LastName_FirstName_MAL1471_Phase2.pdf (or .docx)

Grading Rubric (100 points)

Criterion	Marks	Excellent (90-100%)	Proficient (70-89%)	Developing (50-69%)	Insufficient (<50%)
Completeness of data inventory	25	All major data sources in the VSM are identified and documented (format, owner, storage, quality). Goes beyond obvious sources.	Major sources identified. Some documentation of format and quality.	Data sources listed but with minimal detail.	Absent, generic, or fabricated.
Quality of data gap analysis	30	Gaps are specific, grounded in the VSM, and linked to concrete ML problems. Explains what capturing each gap would enable.	Gaps are real and reasonably specific. Some connection to ML opportunity.	Gaps identified but vague or not clearly linked to ML applications.	No gap analysis, or generic statements without substance.
Technical data understanding	20	Demonstrates understanding of data formats, storage systems, quality issues (missing values, volume, completeness). Applies course EDA vocabulary.	Some technical understanding of data characteristics. Mostly accurate.	Limited technical detail. Largely descriptive rather than analytical.	No technical understanding evident.
Prioritisation and justification	15	Clear, well-reasoned prioritisation. Explains why this source or gap over others: ML value, data availability, feasibility, business impact.	Reasonable prioritisation with some justification.	Priority stated but not well justified.	No prioritisation or arbitrary choice.
Presentation & clarity	10	Well-structured report. Tables are clear. Arguments flow logically.	Mostly clear and readable.	Acceptable but with organisational issues.	Unclear, poorly structured, or incomplete.

Warning

Everything you submit must be verifiable.

Your data landscape must describe the actual information environment of your internship organisation. Your supervisor should be able to read your Phase 2 submission and recognise the data sources and gaps you have identified. The instructor may contact your supervisor to verify this. Analysis of fabricated or invented data sources is an academic integrity violation.